



Hardware-in-the-Loop CPS Security Architecture for Microgrid Monitoring and Control Applications

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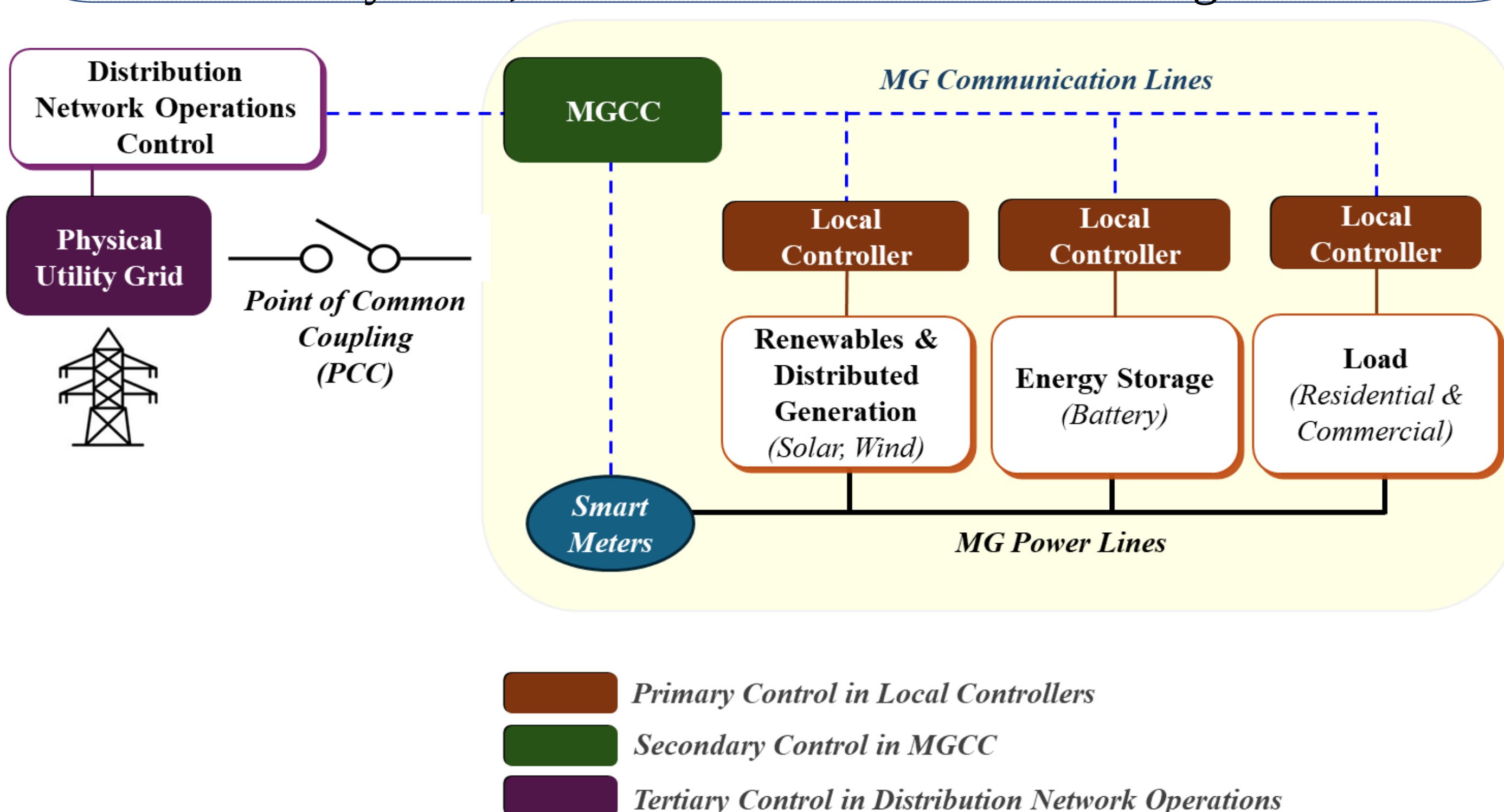
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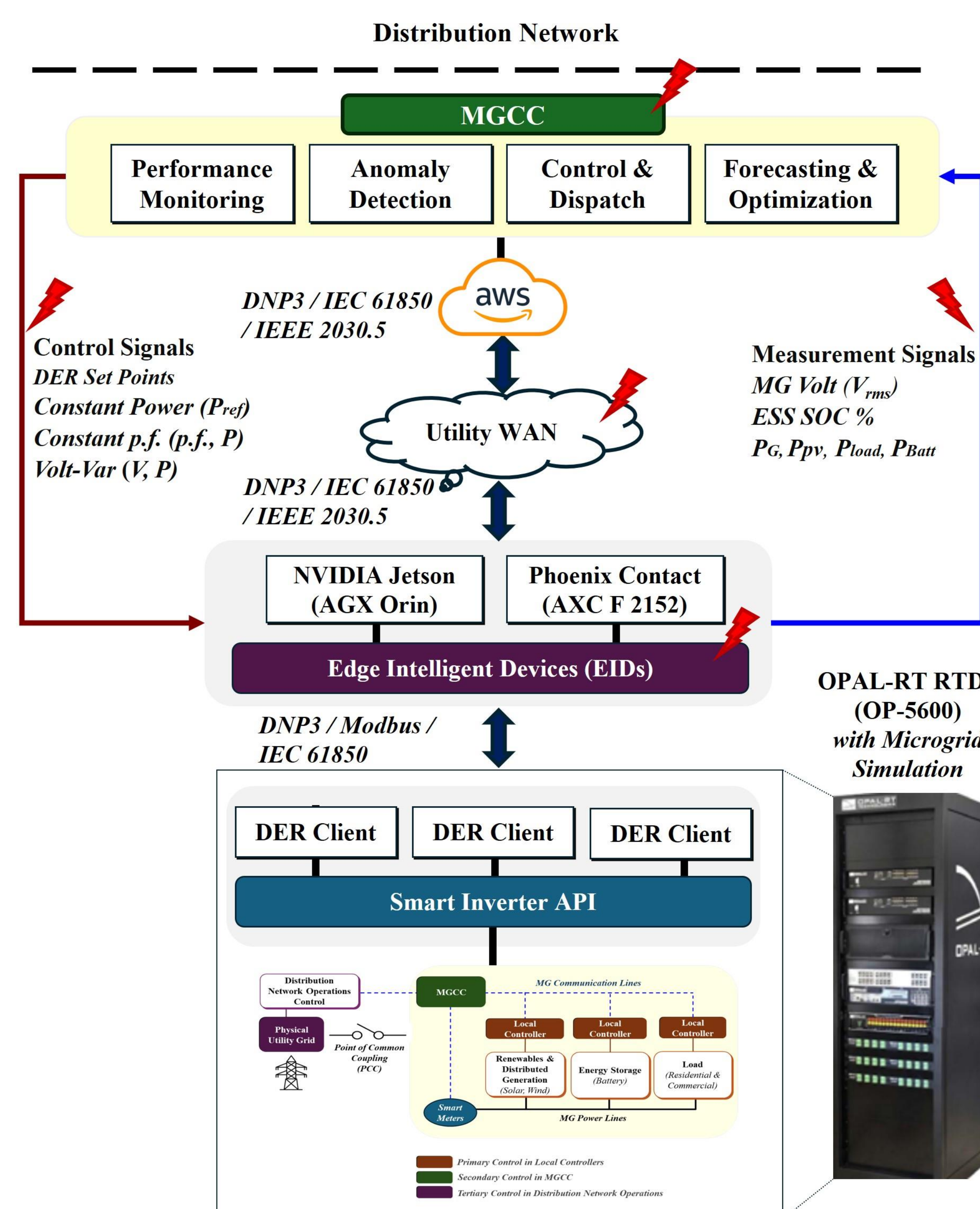
Microgrid Architecture

Research Objective:

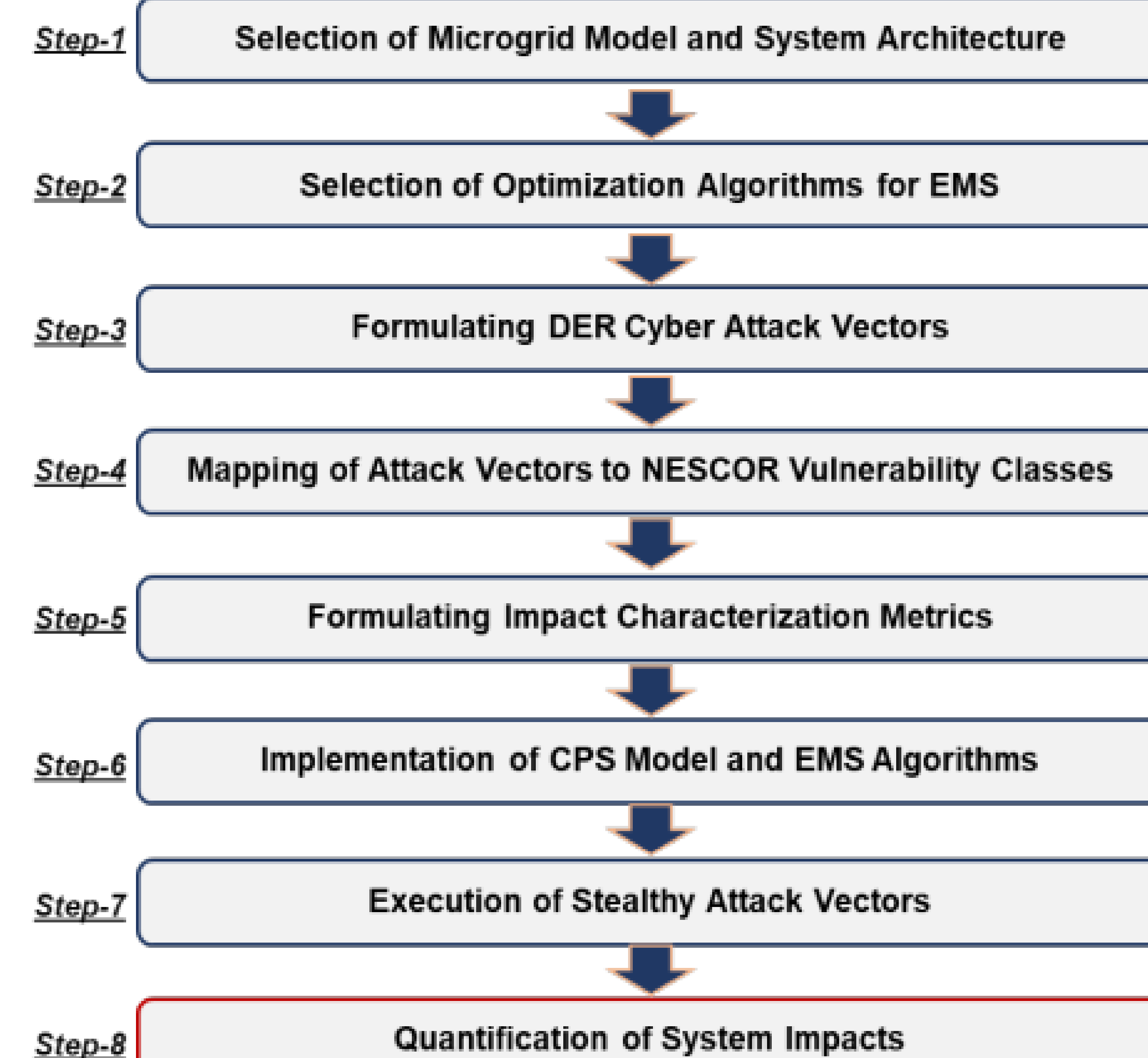
- HIL-CPS-testbed setup of a microgrid is proposed considering the IEEE 1547 standard with industry-grade hardware and software components.
- NESCOR failure scenarios are mapped with the formulated attack vectors and utilize the CPS testbed for high-fidelity cyber attack impact analysis for DER-based microgrid.
- The proposed architecture is helpful for the development and evaluation of real-time anomaly detection & mitigation solutions for distribution systems, which facilitates MG monitoring and control.



Proposed HIL-CPS Testbed Architecture

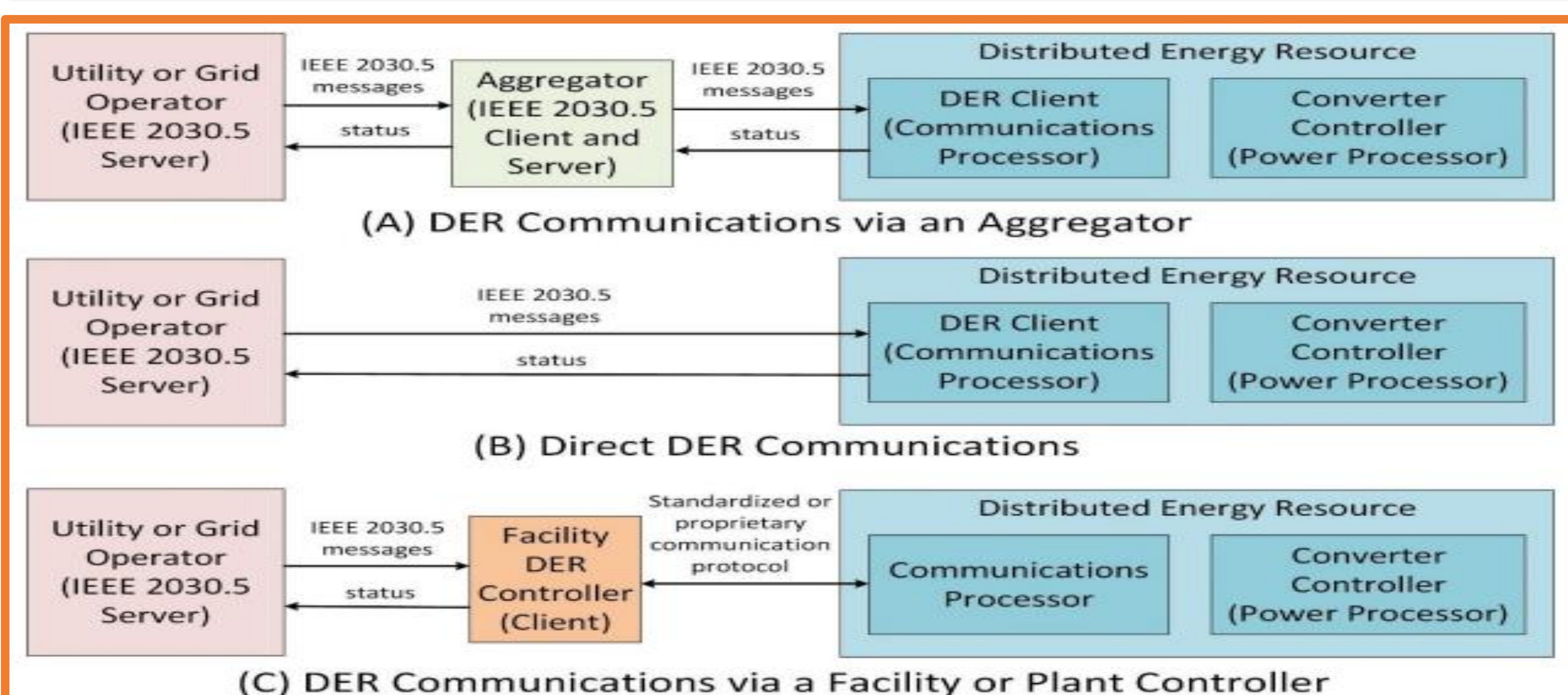


Proposed Cyber-attack Impact Analysis Methodology

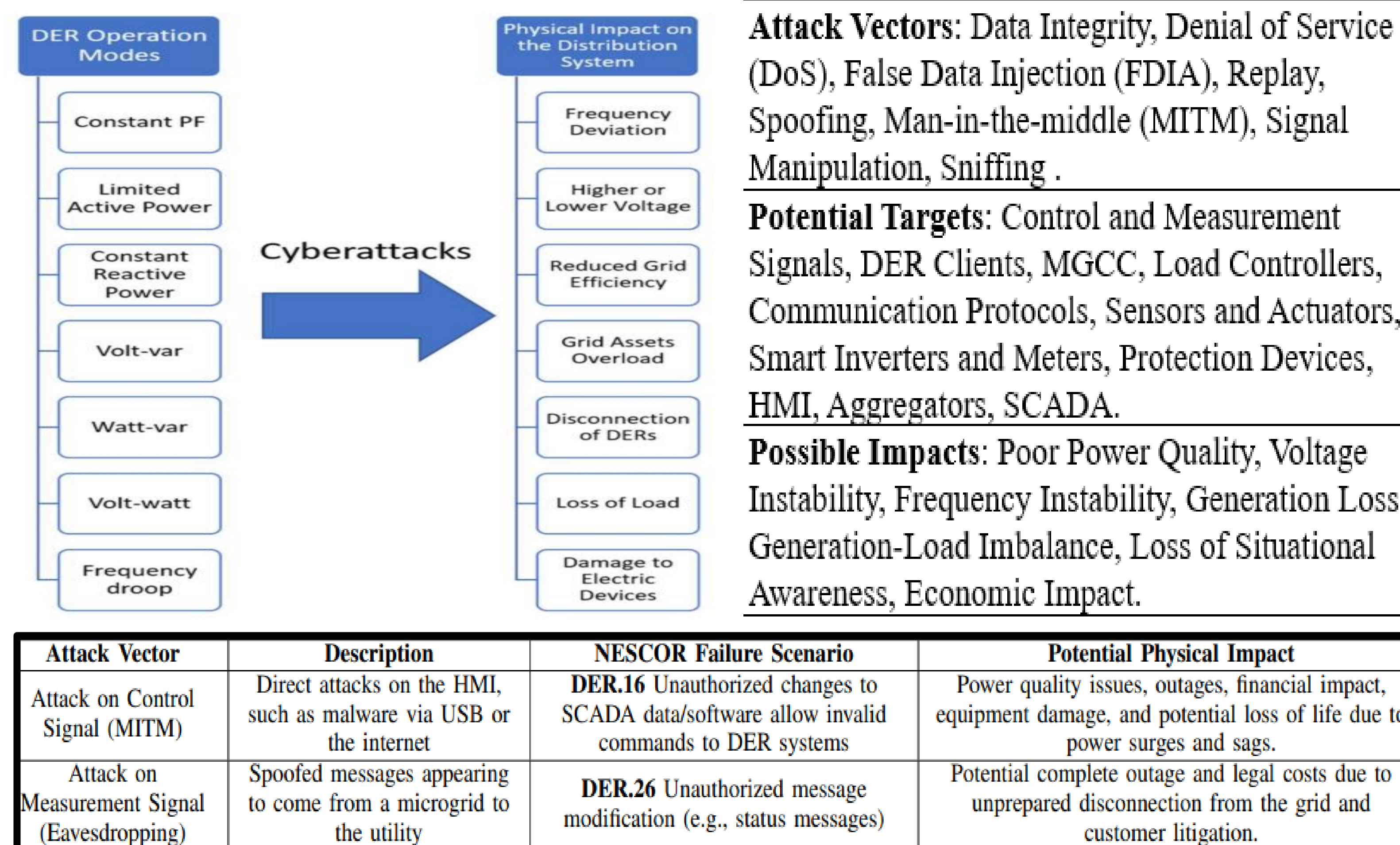


Vulnerabilities of DER Communication Protocols

Protocol	MITM	Replay	DoS	Eavesdropping	Spoofing	Data Modification
SunSpec Modbus	X	X	X	X	X	X
IEEE 1815 (DNP3)	X	X	X	X	X	X
IEEE 2030.5 (SEP2)			X			X



Potential Attack Vectors, Impacts and NESCOR Mapping



Proposed Metrics for Impact Analysis

Operational Cost

$$\min C_T = \sum_{i=0}^N C_G(i) \times P_G(i)$$

Problem constraints,

$$\sum_{i=1}^N P_G(t) + P_{PV}(t) - P_{load}(t) = \pm P_{Batt}(t)$$

$$SOC_{min} \leq SOC(t) \leq SOC_{max}$$

$$E_{batt_min} \leq E_{batt}(t) \leq E_{batt_max}$$

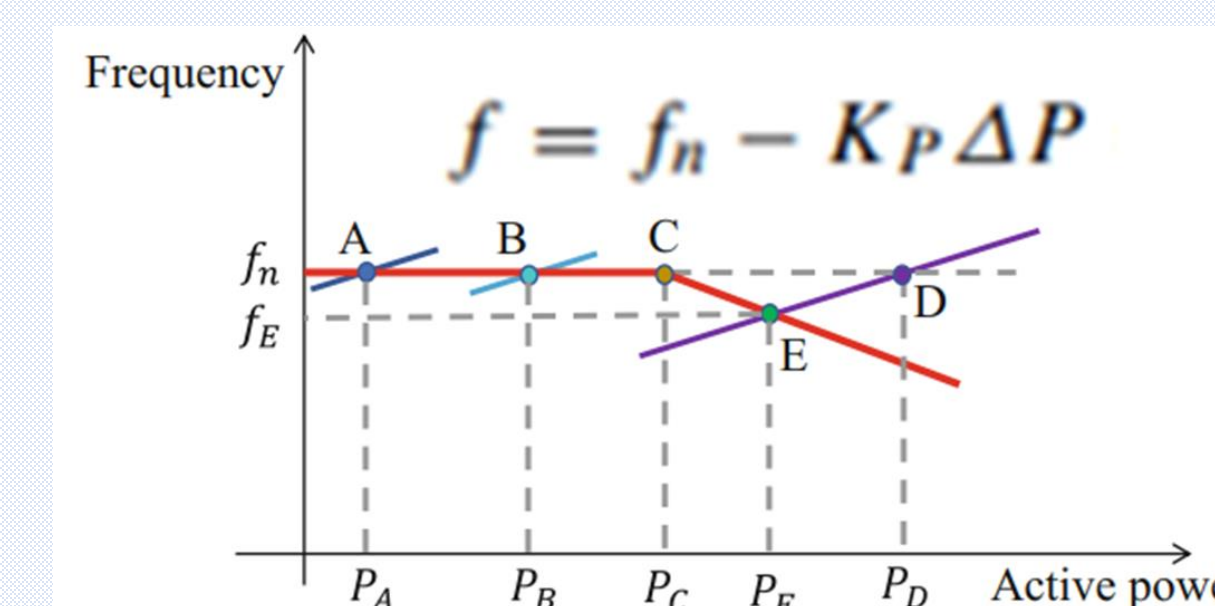
Where,

$$E_{batt_min} = (1 - DOD) \times E_{batt_max}$$

Microgrid's bus Voltage

$$V = V_n - K_Q * \Delta Q$$

Microgrid's System Frequency

$$f = f_n - K_P * \Delta P$$


Reference:

- [1] Recommended Functionalities for Improving Cybersecurity of Distributed Energy Resources <https://www.nrel.gov/docs/ty20osti/74895>
- [2] Sandia National Laboratories <https://sunspec.org/wp-content/uploads/2017/08/DERCyberPrimer-DraftforReview>